

Case 2: a) $f(x) = 3x_1 - 5x_3 + 2x_4 - 3x_5 \rightarrow \text{Max}$

$$\begin{cases} -x_2 + x_4 \leq -2 & y_1 \geq 0 \quad (1) \\ 3x_1 + 2x_2 - x_3 + x_5 \geq 30 & y_2 \leq 0 \quad (2) \\ x_1 + 3x_2 + x_3 - x_4 = 20 & y_3 \in \mathbb{R} \\ -7x_1 - x_3 + 2x_5 \geq -10 & y_4 \leq 0 \quad (3) \end{cases}$$

$$x_1 \leq 0, x_3 \geq 0, x_4 \geq 0$$

giới

$$g(y) = -2y_1 + 30y_2 + 20y_3 - 10y_4 \rightarrow \text{Min}$$

$$\begin{cases} 3y_2 + y_3 - 7y_4 \leq 3 & x_1 \leq 0 \quad (4) \\ -y_1 + 2y_2 + 3y_3 = 0 & x_2 \in \mathbb{R} \\ -y_2 + y_3 - y_4 \geq -5 & x_3 \geq 0 \quad (5) \\ y_1 - y_3 \geq 2 & x_4 \geq 0 \quad (6) \\ y_2 + 2y_4 = -1 & x_5 \in \mathbb{R} \end{cases}$$

$$y_1 \geq 0, y_2 \leq 0, y_3 \in \mathbb{R}, y_4 \leq 0$$

Có 6 cặp ràng buộc đối ngẫu: (1), ..., (6)